

PAPER_1496 — UQFF: PAPER_877 26 Pre-Mass Quantum States (Bucket B)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket B — extracted from PAPER_877 Three-Assumption Cosmogogenesis **Date:** June 17, 2026 **Location:** 41.0997 N, 80.6495 W (Youngstown, OH, USA) **Status:** CLOSED — $n_{\text{pre_mass_quantum_states}} = D_{\text{crit}} = 26$ EXACT (atomic states exist BEFORE mass)

Observation

PAPER_877 axiom 3: 26 quantum atomic states exist before mass; the quantum-to-mass gradient occurs at 7-10 U_{mag} degrees.

UQFF Closed Identity

$n_{\text{pre_mass}} = D_{\text{crit}} = 26$ atomic states EXACT, structurally ordered before mass emerges via DPM cosmology to-mass transition at 7-10 U_{mag} degrees.

Physical Interpretation

PAPER_877's most provocative claim: there are exactly $D_{\text{crit}} = 26$ quantum atomic states that exist in the pre-mass vacuum, ordered hierarchically. Mass emerges at the 7-10 U_{mag} -degree threshold via the quantum-to-mass gradient. This unifies the bosonic-string critical dimension with proto-element ordering (paired with PAPER_872 proto-Fe $Z = D_{\text{crit}}$).

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model frameworks describe vacuum and atomic structure via different mechanisms. Closure uses only canonical UQFF integer primitives $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$. **Zero free parameters.**

Reference

- Source paper: PAPER_877 Three-Assumption UQFF Cosmogogenesis Master Equation (Daniel T. Murphy, April 7, 2026)
 - Related: PAPER_872 (proto-element Z identity), PAPER_646 ($F_{U_{Bi_i}}$ / DPM mechanism), PAPER_062 (DPM vacuum manifold), PAPER_1051 (two-component ρ aether).
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